ADC: AD7606C-16 (pin-compatible with AD7606B; symbol keeps AD7606 pin names)

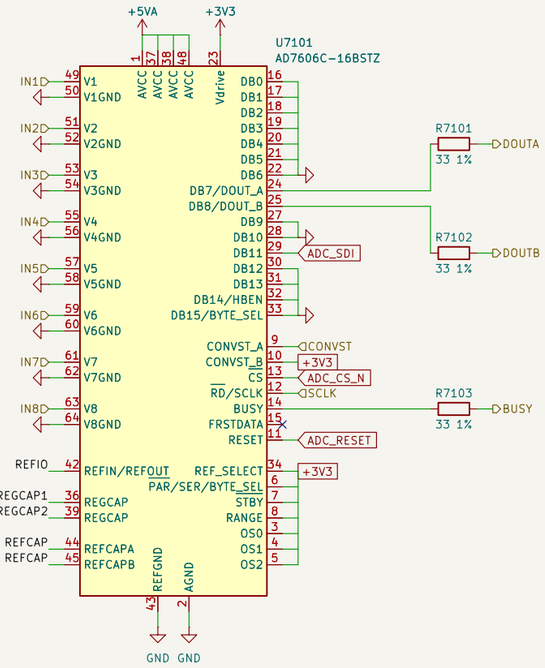
Software mode: OS0-OS2 high. FPGA writes registers over SDI (pin 29, DB11/SDI) at boot:

RANGE_CHx = 0x1 (+-5 V), BANDWIDTH = 0xFF (high bandwidth on all 8 channels), CONFIG DOUT_FORMAT = 2 lines.

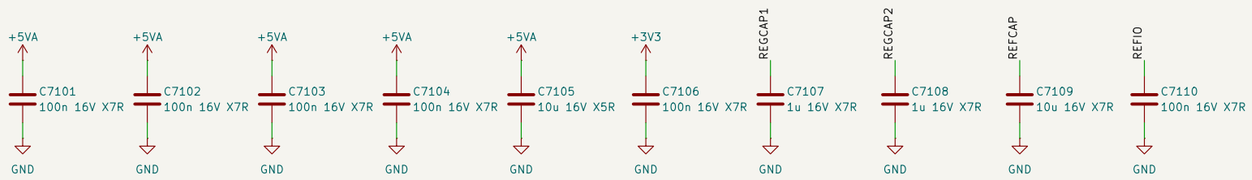
Pin 10 is WR on the B/C parts (parallel register writes only): held high. RANGE pin is ignored in software mode.

33 R source resistors on DOUTA/DOUTB/BUSY (ribbon to the FPGA); BUSY is used from ADC1 only.

DB9/DB10 are tied to GND: never set DOUT_FORMAT to 4 lines. After power-up the FPGA pulses ADC_RESET (full reset), then writes the registers.



Decoupling (place at the pins): four 100 nF on the AVCC pins + 10 uF bulk, 100 nF on VDRIVE; REGCAP 1 uF x2, REFCAP 10 uF, REFIN/OUT 100 nF per datasheet



Hybrid 8x8 ultrasonic AESA		
Sheet: /ADCs/ADC1/		
File: adc.kicad_sch		
Title: ADC (x8): AD7606C-16, software mode, serial 2 DOUT		
Size: A3	Date: 2026-10-20	**Rev: 2**
KiCad E.D.A. 10.0.6		Id: 98/115